Privacy For Sale

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Motivation

A central question in privacy law is if/how to regulate "**third-party data sharing**"

- $\sim$ Firms collect consumer data as by-product of ordinary transactions
- $\sim$ Firms may later sell this data to third parties: advertisers, brokers, etc.

This practice can lower prices and generate surplus, but also impose privacy costs on consumers

Motivation

Policymakers face two broad approaches to regulating **third-party data sharing**:

- Restricting it — including with outright bans
- "Strengthening property rights," which enables firms to make enforceable data-use promises

This Paper

How does regulation of **third-party data sharing** affect how products are sold, how data is monetized, and social welfare?

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Seller

Consumer

Third Party

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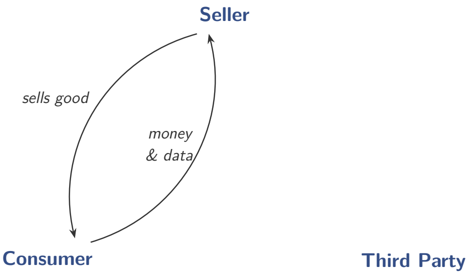

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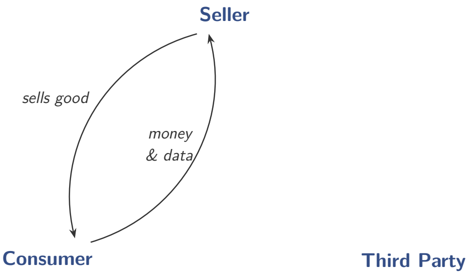

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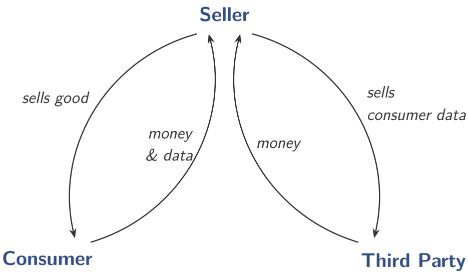

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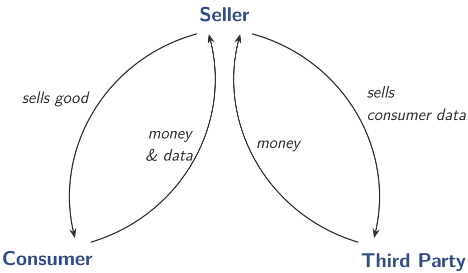

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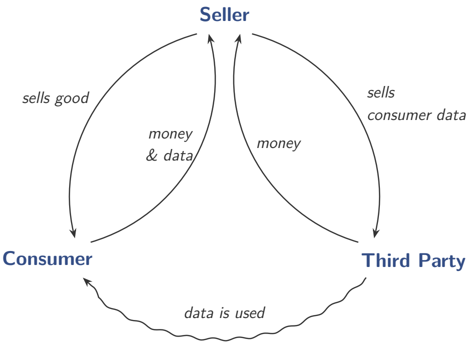

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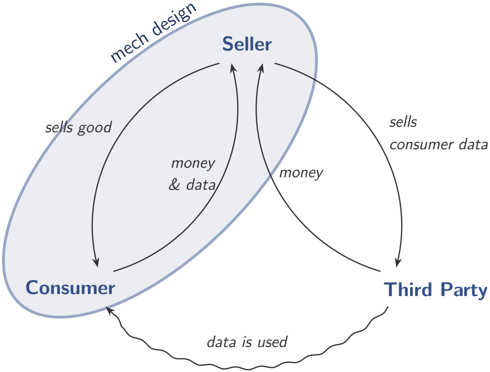

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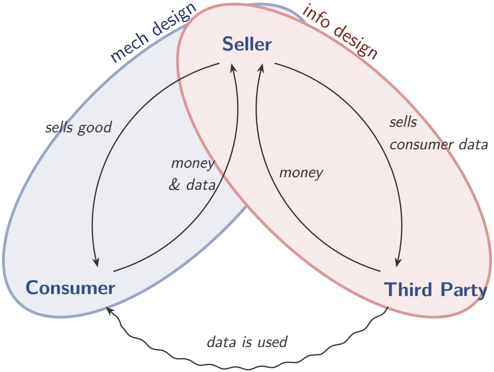

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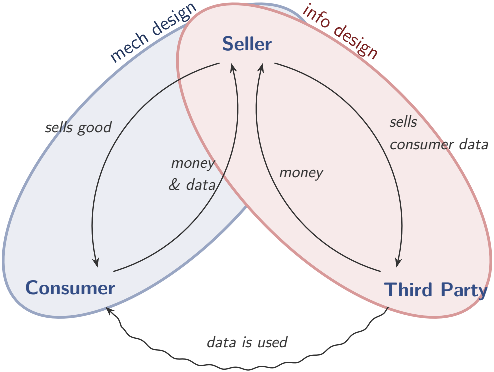

Tradeoff: Mechanism shapes data sale; data sale shapes mechanism

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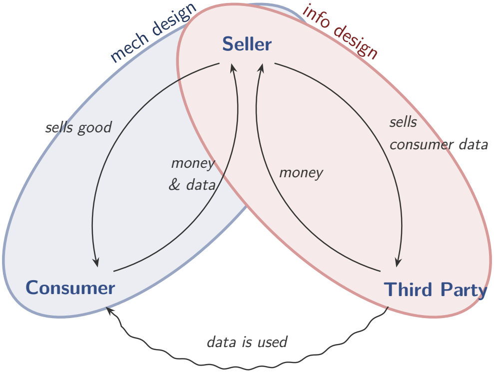

Policy Variations: Whether or not seller can commit to data sales

Main Results

We uncover importance of novel stuctural parameter: the **privacy** β

- $\sim$ It governs whether consumers with higher WTP for seller's good also have higher ($\beta > 0$) or lower ($\beta < 0$) WTP for privacy

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When $\boxed{\beta > 0}$

- Third-party data sharing is **coarse**, and more data use hurts consumers
- Optimal policy tries to restrict third-party data sharing

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When $\boxed{\beta > 0}$

- Third-party data sharing is **coarse**, and more data use hurts consumers
- Optimal policy tries to restrict third-party data sharing

When $\boxed{\beta \leq 0}$

- Third-party data sharing is **granular**, and more data use benefits consumers
- Optimal policy tries to encourage third-party data sharing

Related Literature

We contribute to growing literature on the **economics of data and privacy**

Acquisti, Taylor, Wagman (2016), Bergemann, Bonatti, Smolin (2018), Bergemann and Bonatti (2019), Goldfarb, Tucker (2023), Galperti, Levkun, Perego (2024), Baley and Veldkamp (2025), Galperti, Liu, Perego (2026)

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We also contribute to literature on **info sharing in mechanism design**

Calzolari and Pavan '06, Dworczak '20	conditions for optimality of privacy
Rayo '13, Doval and Skreta '25, Lu '25	Mussa-Rosen settings
Dworczak '17, Corrao '23, Rivera Mora '25	submod vs supmod
Eilat, Eliaz, Mu '21, Krahmer and Strausz '23	privacy in mechanism design

model

Model: First Period

The seller has a single indivisible good for sale

$\theta \in [0,1]$ is the consumer's valuation for this good, distributed with CDF F

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- If $x(m) = 0$, product is not allocated, game ends, payoffs are zero
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In the latter case, we refer to message m as the consumer's **data record**

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If record m is sold, the third party

- $\sim$ gains access to the consumer and interacts with her by taking action $a \in \mathbb{R}$
- $\sim$ we interpret action a as **data use** — e.g., intensity of third-party advertising

Data Policies

We compare three alternative policy environments:

- **Data "Anarchy"**. The seller **cannot commit** to the second-period data-pricing strategy $p : M \to \mathbb{R}$ (lassez faire)

- **Data "Market"**. The seller **can commit** to the second-period data-pricing strategy $p : M \to \mathbb{R}$ ("strengthen property rights")

- **Data "Ban"**. The seller **cannot sell** consumer data to the third party (restrict use)

Payoffs: Seller

The **seller**'s payoff is

$$t(m) + p(m)$$

the revenue from selling **the good** to consumer and **the data** to third party

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For example, $v(a, \theta) = a\theta - \frac{1}{2}a^2$

Payoffs: Consumer

Type-θ consumer's payoff from reporting m when third party's action is a:

$$x(m)\Big(\theta - a \cdot \ell(\theta)\Big) - t(m),$$

$\ell(\theta)$ is type-θ's marginal disutility from data use $\rightsquigarrow$ **privacy loss** (1-dim type)

Instrumental (as opposed to intrinsic) preference for privacy

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For this talk, I'll focus on **affine** and **positive** privacy loss function

$$\boxed{\ell(\theta) = \alpha + \beta\theta \geq 0}$$

I refer to the slope parameter as the "**privacy** β"

All results extend to any monotone and convex ℓ

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If β negative: Consumers who value good more are **less** averse to data use

i.e., there is a **negative** correlation between valuation and privacy loss

Example: Third party advertises a complementary product

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implementable outcomes

Outcomes

Let an *outcome* (or direct mechanism) be described by (x, t, a) where

- $x : \Theta \to \{0, 1\}$ is the allocation rule
- $t : \Theta \to \mathbb{R}$ is the monetary transfer from consumer
- $a : \Theta \to A$ is the recommendation rule / information structure for TP

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An outcome (x, t, a) is

- *individually rational* (IR) if $U(\theta) \geq 0$ for all θ
- *incentive compatible* (IC) if $U(\theta) \geq u(\theta, \theta')$ for all θ, θ'
- *obedient* (OB) if $\tilde{a} \in \arg\max_{a'} \mathbb{E}\big(v(\theta, a') \mid a(\theta) = \tilde{a}\big)$ for all $\tilde{a} > 0$

Characterization of Implementable Outcomes

An outcome is *implementable* in a policy regime if it can be induced by an equilibrium for that regime

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Remark 1

An outcome (x, t, a) is implementable in

1. Data Market regime iff it satisfies IR, IC, and OB
2. Data Anarchy regime iff it satisfies IR, IC, OB, and additionally $a(\theta) > 0$ for all θ such that $x(\theta) = 1$
3. Data Ban regime iff it satisfies IR, IC, OB, and additionally $a(\theta) = 0$ for all θ

1. Seller can commit to data-pricing strategy $\Rightarrow$ Standard application of RP

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2. Seller can't commit to data-pricing strategy: Each record has positive value for TP $\Rightarrow$ it will be sold $\Rightarrow a(\theta) > 0$ if $x(\theta) = 1$

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3. Seller can't sell records $\Rightarrow a = 0$ (OB is vacuous)

the role of privacy β

How Privacy Preferences Shape Data Monetization

To collect consumer data, the seller must sell product versions that credibly differ in their privacy attributes

Question: To what extent can seller monetize the data and how?

E.g, are data records coarse (i.e., revealing limited information), or granular (i.e., revealing a lot of information)?

The answer depends crucially on the sign of the privacy β

How Privacy Preferences Shape Data Monetization

Proposition

Consider any policy regime in which data sales are permitted:

- If $\boxed{\beta > 0,}$ third-party data sharing is "**coarse**"

  In any implementable outcome, at most one data record is sold; this record pools all types of consumers whose data is sold

- If $\boxed{\beta \leq 0,}$ third-party data sharing can be "**granular**"

  There is an implementable outcome in which a continuum of data records is sold, each revealing a unique consumer type

How Privacy Preferences Shape Data Monetization

Therefore,

- If $\boxed{\beta > 0}$

  Consumers whose data is sold must receive the same ex-post treatment, and are compensated equally, even if the value of their data is different

- If $\boxed{\beta \leq 0}$

  Consumers whose data is sold can receive different ex-post treatment, and thus different compensation

How Privacy Preferences Shape Data Monetization

Intuition: If $\boxed{\beta \geq 0}$

- IC $\Rightarrow$ seller must ensure that $a(\theta)$ is **non-increasing** on the set of trading types.

    High types are more privacy sensitive, so they must not be assigned more data use to prevent deviations

- OB + monotone $a \Rightarrow$ seller must ensure that $a(\theta)$ is **non-decreasing** on the set of trading types

    This is because higher types call for higher actions from third party

Thus, IC and OB are **in conflict** $\rightsquigarrow$ at most one strictly positive recommendation

How Privacy Preferences Shape Data Monetization

Intuition: If $\boxed{\beta \leq 0}$

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  Low types care more about privacy, so they must not be assigned more data use to prevent deviations

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from implementability to equilibria

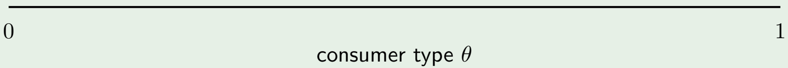
positive β

(higher types are more privacy-sensitive)

Equilibrium Outcomes When $\beta > 0$

Proposition

Fix $\boxed{\beta > 0}$ and a policy regime $i \in \{\mathsf{A}, \mathsf{B}, \mathsf{M}\}$. There exist cutoffs $\theta_g^i \leq \theta_d^i$ such that an equilibrium outcome (x^i, a^i, t^i) is as follows:



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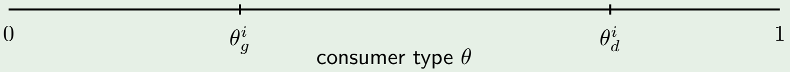

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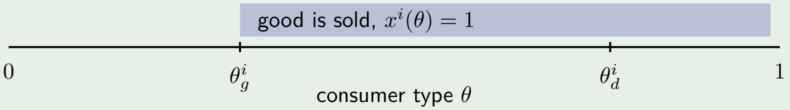

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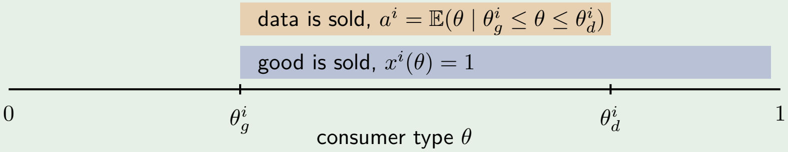

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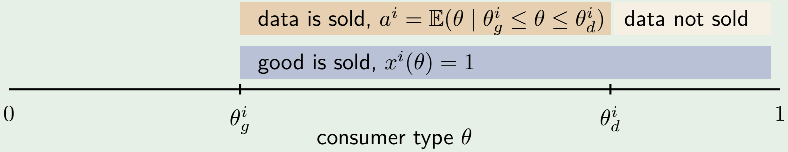

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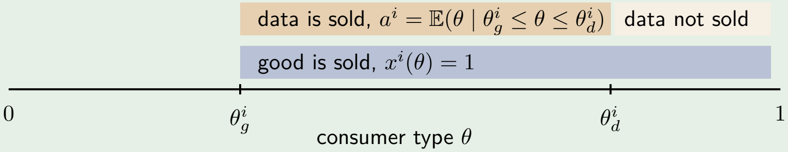

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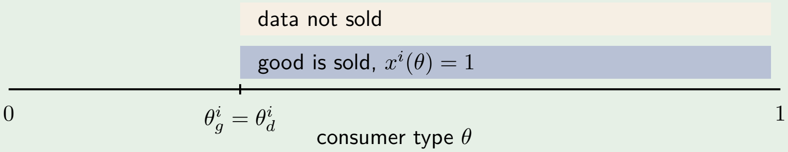

In particular, under **Data Ban**, $\theta_d^{\mathsf{B}} = \theta_g^{\mathsf{B}} \rightsquigarrow$ <u>no</u> data is sold

- No aftermarket $\Rightarrow$ standard monopoly pricing problem
- Simple posted price mechanism

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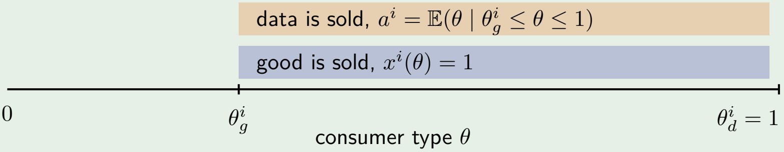

In particular, under **Data Anarchy**, $\theta_d^{\mathsf{A}} = 1 \rightsquigarrow$ all data is sold

- Monetization is coarse; consumers are pooled in same record
- Still posted price. Same treatment, same “compensation”

Equilibrium Outcomes When $\beta > 0$

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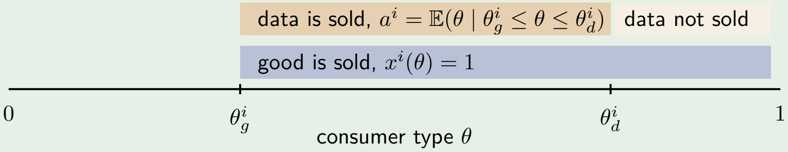


In particular, under **Data Market**, $\theta_d^{\mathsf{M}} \in [\theta_g^{\mathsf{M}}, 1] \rightsquigarrow$ <u>some</u> data is sold

- Two versions of the good are offered: one is private, the other is not
- Private version attracts higher types, who are more privacy sensitive

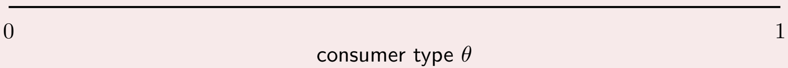
negative β

(higher types are less privacy-sensitive)

Equilibrium Outcomes When $\beta \leq 0$

Proposition

Fix $\boxed{\beta \leq 0}$ and a policy regime $i \in \{\mathsf{A}, \mathsf{B}, \mathsf{M}\}$. There exist cutoffs $\theta_g^i \leq \theta_d^i$ such that an equilibrium outcome (x^i, a^i, t^i) is as follows:



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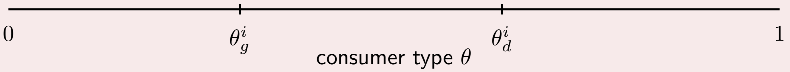

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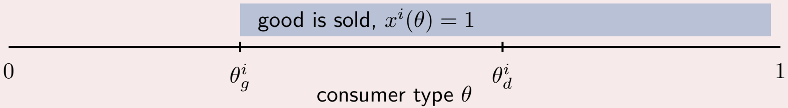

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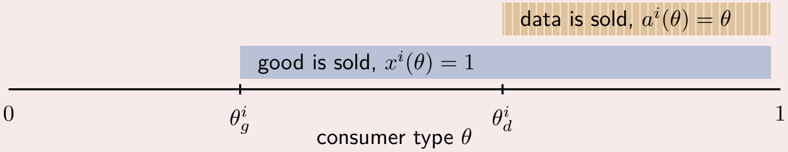

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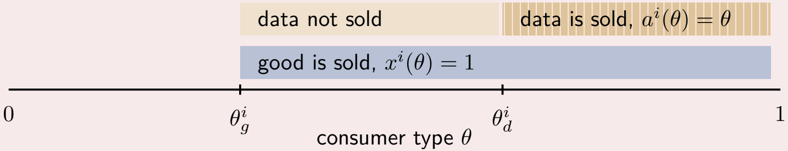

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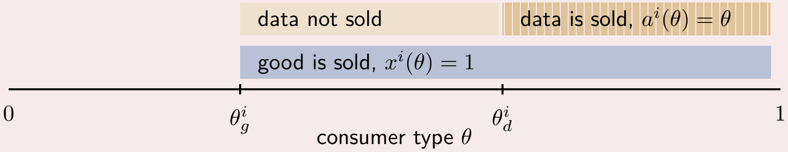

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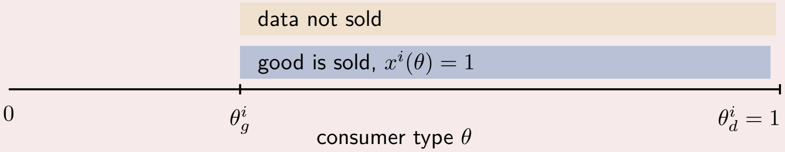

In particular, under **Data Ban**, $\theta_d^{\mathsf{B}} = 1 \rightsquigarrow$ <u>no</u> data is sold

- No aftermarket $\Rightarrow$ standard monopoly pricing problem
- Simple posted price mechanism

Equilibrium Outcomes When $\beta \leq 0$

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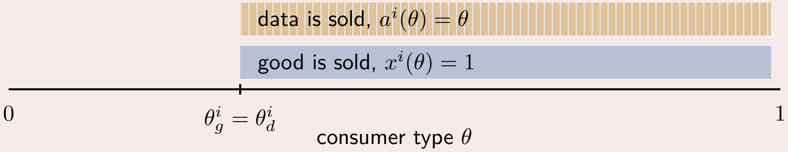


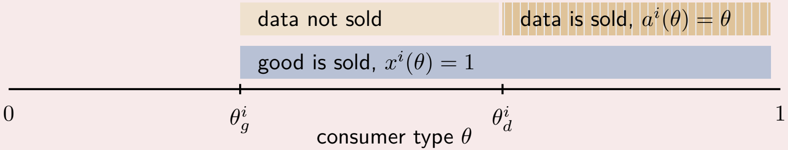
In particular, under **Data Anarchy**, $\theta_d^{\text{A}} = \theta_g^{\text{A}} \rightsquigarrow$ all data is sold

- Monetization is granular; third party learns θ (feasible vs optimal)
- Rich menu: each version with a different expost treatment and comp

Equilibrium Outcomes When $\beta \leq 0$

Proposition

Fix $\boxed{\beta \leq 0}$ and a policy regime $i \in \{\text{A}, \text{B}, \text{M}\}$. There exist cutoffs $\theta_g^i \leq \theta_d^i$ such that an equilibrium outcome (x^i, a^i, t^i) is as follows:



In particular, under **Data Market**, $\theta_d^{\text{M}} \in [\theta_g^{\text{M}}, 1] \rightsquigarrow$ some data is sold

- Two versions of the good are offered: one is private, the other is not
- Nonprivate version attracts higher types, who are less privacy sensitive

Summary

Thus, the sign of privacy β determines

- not only whether data sales are granular or coarse,
- but also how consumers sort across the privacy options, and at what price

Note: Data market allows seller to **unbundle** sale of good from sale of data welfare

How Policies Shape Welfare

What are the welfare implications of these policies?

Seller is weakly better off in Data Market (**Remark 1**)

Third party is indifferent across regimes

What about consumers?

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What about consumers?

Fixing allocation x, does more data use benefit or hurt consumers?

How Data Monetization Shapes Consumers Welfare

Fixing allocation x, does more data use benefit or hurt consumers?

Recall consumers dislike data use by assumption (since $\ell(\theta) \geq 0$)

Yet, *in equilibrium*, outcomes with "more data use" are not necessarily worse for consumers, since transfers adjust when data use changes

The question is whether the transfer adjustment more than offsets or only partially offsets the resulting privacy loss

How Data Monetization Shapes Consumers Welfare

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Recall consumers dislike data use by assumption (since $\ell(\theta) \geq 0$)

Yet, *in equilibrium*, outcomes with "more data use" are not necessarily worse for consumers, since transfers adjust when data use changes

The question is whether the transfer adjustment more than offsets or only partially offsets the resulting privacy loss

The answer depends again on the sign of the privacy β

How Data Monetization Shapes Consumers Welfare

Proposition

Among implementable outcomes with the same allocation rule x:

- If $\boxed{\beta \geq 0,}$ higher data use **decreases** consumer payoff $U(\theta)$ for all θ
- If $\boxed{\beta \leq 0,}$ higher data use **increases** consumer payoff $U(\theta)$ for all θ

How Data Monetization Shapes Consumers Welfare

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Among implementable outcomes with the same allocation rule x:

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Economic Implication: Same policy will have different effect depending on β

- A policy that discourages data use (i.e., Ban) may benefit consumers when $\beta > 0$ but hurt them when $\beta \leq 0$
- A policy that encourages data use (i.e., Anarchy) may benefit consumers when $\beta \leq 0$ but hurt them when $\beta > 0$

Welfare Implications

The complication is that a policy change will also change allocation x

Welfare Implications

Proposition

If $\boxed{\beta \leq 0,}$

- Data Market Pareto dominates Data Ban
- There exists $\tilde{\theta}$ s.t. types below $\tilde{\theta}$ prefer Market to Anarchy, while types above $\tilde{\theta}$ prefer Anarchy to Market

If $\boxed{\beta > 0,}$

- Data Market Pareto dominates Anarchy (when β sufficiently small)
- There exists $\tilde{\theta}$ s.t. types below $\tilde{\theta}$ prefer Market to Ban, while types above $\tilde{\theta}$ prefer Ban to Market

Summary of Welfare Implications

Thus, a data market always Pareto dominates either data anarchy or a data ban—with the sign of the privacy β determining which

It is never itself Pareto dominated

In this sense, establishing a data market emerges as an attractive policy choice, performing well across environments

Instead, data ban and data anarchy can be effective in some circumstances but counterproductive in others

conclusion

Conclusion

A central concern of privacy law is how to regulate third-party data sharing

We study a three-party model combining mech- and info-design components

We uncover importance of a new structural parameter: the privacy β

- Its sign shapes (1) how the seller can monetize data; (2) Whether more data use benefits or hurts consumers; (3) Which data policy is Pareto optimal

Optimal regulation is nuanced, but Data Markets are never dominated and always outperform either Ban or Anarchy.

Privacy For Sale

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Thanks!